OBSERVING TIME APPLICATION

Differential photometry of a short-period eclipsing binary

1. PROPOSAL SUMMARY

PROPOSAL TITLE
Differential photometry of a short-period eclipsing binary

CATEGORY
Stellar Astrophysics

PROGRAM TYPE
Regular Proposal (RP)

EXECUTION MODE
Queue / Service Observing

OBSERVING MODE
Photometry

INSTRUMENT
ASI2600MM Pro + RGB + OAG-ASI120MM Mini

ASSOCIATED ACADEMIC WORK
PhD thesis

STUDENT / ADVISOR
Example Student / Example Advisor

2. ABSTRACT

We propose time-series photometry of a bright eclipsing binary to measure the eclipse depth, refine the time of minimum light and assess short-term variability. Observations in the G and R filters will provide a continuous light curve with a cadence below one minute. The resulting calibrated differential photometry will be compared with published ephemerides and modelled to constrain the eclipse morphology.

3. PRINCIPAL INVESTIGATOR

NAME
Example Student

INSTITUTION
INPE

E-MAIL
student@example.edu

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+55 12 0000-0000

ORCID
Not provided

4. CO-INVESTIGATORS

Name	Institution	E-mail	Phone
Example Advisor	INPE	advisor@example.edu	—

OBSERVING REQUEST AND TECHNICAL CONFIGURATION

Photometry · ASI2600MM Pro + RGB + OAG-ASI120MM Mini

5. TIME REQUEST AND SCHEDULING

TOTAL REQUESTED TIME
4 h

MINIMUM USEFUL TIME
2.50 h

PRIMARY DATES
13 Aug 2026

BACKUP DATES
20 Aug 2026, 21 Aug 2026

PREFERRED INTERVAL
18:30–23:00

AFTER MIDNIGHT
No

MOON CONSTRAINT
Any lunar phase

WEATHER / AIRMASS
Clear; <= 5 arcsec; airmass 2.0

DISTRIBUTION
A single continuous four-hour block is preferred.

UNUSABLE DATES
None specified

6. INSTRUMENT SETUP

INSTRUMENT
ASI2600MM Pro + RGB + OAG-ASI120MM Mini

FILTERS
G, R

BINNING
2 × 2

READOUT
1.11 s/frame

GAIN / OFFSET
100 / Instrument standard

DETECTOR TEMPERATURE
-10 °C

ROI / SUBFRAME
Full frame

DESIRED CADENCE
45 s

DITHERING
No

CALIBRATIONS
Bias/dark + flats in G and R

COMPARISON STARS
Ensemble differential photometry with at least three non-variable comparison stars.

7. TARGET LIST

Target	RA	Dec	Mag/Band	Type	Exp.	N×V	Comments
Example Binary	18:00:00	-20:00:00	10.5 V	Science	30s	300×1	Continuous monitoring through eclipse

SCIENTIFIC JUSTIFICATION

Eclipsing binaries provide direct constraints on stellar radii, orbital geometry and stellar evolution. The selected target has a well-established short period but its published ephemeris has accumulated uncertainty. A densely sampled eclipse obtained in two filters will measure a new time of minimum light and test whether the eclipse depth is wavelength dependent.

EXPERIMENTAL DESIGN

The target will be monitored continuously before, during and after the predicted eclipse. Alternating G and R exposures will provide colour information while maintaining a cadence below one minute. The field contains several comparison stars of similar brightness. Bias/dark frames and flats in both filters will be obtained with the same detector configuration.

TECHNICAL JUSTIFICATION

Thirty-second integrations keep the target and comparison stars within the linear detector regime while providing sufficient signal-to-noise. Including the 1.11 s readout, the expected cadence is approximately 31 s per frame. Four hours cover the baseline and full eclipse, while 2.5 hours constitute the minimum useful allocation.

DATA REDUCTION AND ANALYSIS

Images will be bias/dark corrected and flat-fielded. Differential aperture photometry will be extracted for the target and an ensemble of comparison stars. Time stamps will be converted to a consistent barycentric system. The light curve will be fitted with an eclipse model, and the time of minimum and its uncertainty will be estimated by resampling.

EXPECTED RESULTS AND PUBLICATION POTENTIAL

A calibrated two-band light curve, a new eclipse timing measurement and a comparison with the published ephemeris.

BACKUP PROJECT AND RISK MITIGATION

If transparency is variable but the target remains visible, differential photometry in G will be prioritised. If the primary target is unavailable, a second bright variable in the same sky region will be observed using the same configuration.

REFERENCES

Example et al. 2024, AJ, 000, 1